Pranay Ugale

Machine Learning Engineer **|** B.Tech. CSE (Data Science) **|** Remote-Ready

pranayugale31@gmail.com | +91 8668341326 | linkedin.com/in/pranayugale | github.com/pranay-3108

Machine Learning enthusiast pursuing B.Tech. in CSE (Data Science), seeking an ML internship to apply hands-on experience in model development, computer vision, and Python-based ML workflows. Proven ability to work independently on real-world datasets, build clean data pipelines, and implement end-to-end ML solutions.

TECHNICAL SKILLS

ML: Supervised & Unsupervised Learning, Classification, Regression, Model Evaluation
DL / CV: Neural Networks, CNN, U-Net, Transfer Learning, Image Segmentation, Pattern Recognition
Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Tools: Python, SQL, Git, GitHub, Jupyter Notebook, Google Colab, Kaggle
Soft Skills: Analytical Thinking, Problem Solving, Independent Remote Work, Documentation

EDUCATION

B.Tech. in CSE (Data Science) | Gaikwad Patil College of Engineering 2023 – Present

GPA: 8.67 / 10 | Highest Grade: 9.19

PROJECTS

Chest X-ray Pneumothorax Segmentation | U-Net · PyTorch · SIIM-ACR · Kaggle 2024 – Present

- Built a U-Net model from scratch in PyTorch for pixel-level lung segmentation on 10,000+ DICOM chest X-rays (~3GB SIIM-ACR dataset).
- Engineered a complete data pipeline: mapped ImageId to file paths, parsed RLE masks from CSV, handled nested folders and negative samples (–1 labels).
- Processed 3,000+ image-mask pairs with zero corrupted samples; verified clean mask overlays confirming accurate disease region detection.
- Evaluated with Dice Coefficient & IoU; iteratively improved submissions on Kaggle leaderboard.

Chessboard State Recognition & FEN Generator | CNN · Computer Vision · Python 2023 – 2024

- Built an end-to-end pipeline using a custom CNN to detect and classify chess pieces from board images with high accuracy.
- Self-labeled the entire training dataset; model demonstrated robustness to blur, glare, and low-quality images through data augmentation.
- Automated FEN notation generation by integrating piece classification and structured output — delivering a fully working computer vision application.

RELEVANT ACTIVITIES

- **Kaggle Competitor:** Participated in image segmentation and tabular data competitions; practised feature engineering, model comparison, and experiment tracking.
- **Daily DSA Practice:** Consistent problem-solving in Python to strengthen algorithmic thinking and coding proficiency.
- **Open-Source Contributor:** Submitted pull requests and bug fixes to public repositories on GitHub.

ACHIEVEMENTS

- Scored 9.19 GPA in the highest academic semester during B.Tech. CSE (Data Science) program.
- Completed a Kaggle competition submission with iterative model improvements on a 10,000+ image medical dataset.
- Independently built and documented two full ML projects covering data acquisition, preprocessing, modeling, evaluation, and deployment.

LANGUAGES

English (Professional) Hindi (Native) Marathi (Native)